



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.NOS.1

# Divide Mixed Numbers

Lesson 6-10 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can divide mixed numbers by first changing them to improper fractions.

**Language Objective:** I can explain my steps using the words mixed number, improper fraction, convert, reciprocal, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mixed Number

Improper Fraction

Convert

Simplify

Reciprocal



Tap any word to see its meaning and a picture.

1

A number with a whole number and a fraction, like  $2\frac{1}{3}$ , is a

2

A fraction whose numerator is greater than or equal to its denominator, like  $\frac{7}{3}$ , is an

3

Changing a mixed number into an improper fraction is to

 it.
 

4

Writing a fraction with smaller numbers that names the same amount, like  $\frac{4}{8} = \frac{1}{2}$ , is to

 it.
 

5

To divide by a fraction, I multiply by its

## Write About the Math

### How does dividing mixed numbers help the detective?

*¿Cómo ayuda dividir números mixtos a la detective?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Mixed Number**  
*Número mixto*

☐ **Improper Fraction**  
*Fracción impropia*

☐ **Convert**  
*Convertir*

☐ **Simplify**  
*Simplificar*

☐ **Reciprocal**  
*Recíproco*

**Model:** A mixed number must first become an improper fraction so I can flip and multiply. — Students explain that the mixed number must be converted to an improper fraction ( $7/2$ ) before applying Keep, Change, Flip.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The detective can cut \_\_\_\_ pieces because  $5 \frac{1}{4} \div \frac{3}{4} = \frac{21}{4} \times \frac{4}{3} = \frac{84}{12} = \underline{\hspace{1cm}}$ .
- My answer is \_\_\_\_ because \_\_\_\_.

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.  
*Mi afirmación es \_\_\_\_.*
- My evidence is \_\_\_\_, so \_\_\_\_.  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Mixed Number
2. **Notes 2:** Improper Fraction
3. **Notes 3:** Convert
4. **Notes 4:** Simplify
5. **Notes 5:** Reciprocal
6. **Watch for this mistake:** *A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first — for example, treating  $4\frac{1}{2} \div 1\frac{1}{2}$  as  $(4 \div 1) + (1/2 \div 1/2) = 5$ . Always convert each mixed number to one improper fraction first, then divide using Keep, Change, Flip:  $4\frac{1}{2} \div 1\frac{1}{2} = 9/2 \div 3/2 = 9/2 \times 2/3 = 3$ .*
7. **Try It 1:** B.  $2 - 3 \div 1\frac{1}{2} = 3 \div 3/2 = 3 \times 2/3 = 6/3 = 2$  sections.
8. **Try It 2:** B.  $3 - 7\frac{1}{2} = 15/2$  and  $2\frac{1}{2} = 5/2$ . So  $15/2 \div 5/2 = 15/2 \times 2/5 = 30/10 = 3$  doorways.